San Francisco Bay Regional Water Quality Control Board

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Google Bay View Recycled Water Project
Attn.: Drew Wenzel and Steve Grass
100 Lomax Lane
Mountain View, CA 94043
Sent electronically to dwenzel@google.com
Sent electronically to stevegrass@google.com

Subject: **Notice of Applicability, Water Quality Order WQ 2014-0153-DWQ Google Bay View Campus Recycled Water Treatment System, Mountain View, Santa Clara County**

Dear Messrs. Wenzel and Grass:

This letter serves as the Notice of Applicability (Notice) from the San Francisco Bay Regional Water Quality Control Board (Water Board) that the State Water Resources Control Board's (State Water Board's) General Waste Discharge Requirements for Small Domestic Wastewater Treatment Systems, Order WQ 2014-0153-DWQ (General Order) is applicable to the subject wastewater and stormwater treatment system and recycled water non-potable uses are henceforth regulated under the General Order as described below. A Monitoring and Reporting Program (Monitoring Program) for the Google Bay View Campus recycled water treatment system is included as Attachment B of this Notice, which supersedes the model Monitoring Program in Information Sheet C of the General Order.

Sherwood Design Engineers submitted the *Title 22 Engineering Report*[1] *for the Treatment and Reuse of Wastewater and Stormwater*, dated August 2019, followed by revised versions, in July 2020 and 2021, and on October 27, 2021 (Title 22 Engineering Report), on behalf of Google LLC (Discharger) for the Google Bay View Campus wastewater and stormwater recycled water treatment system. The system is located at the onsite central utility plant at 100 Lomax Lane, Mountain View in Santa Clara County (Facility). Additionally, Google submitted the *Recycled Water System Engineering Report Dual-Plumbed Building* (Dual Plumbed Report), dated December 22, 2017, and revised on November 9, 2018. The *Preliminary Operations and Maintenance Plan and*

[1] A Title 22 Engineering Report is documentation that addresses the production, transmission, and reuse of recycled water and compliance with the requirements of the California Code of Regulations Title 22 (22 CCR or Title 22) division 4, chapter 3 Water Recycling Criteria.

Contingency Plan (O&M and Contingency Plan), dated December 15, 2021, was also submitted to the Water Board and the Division of Drinking Water (DDW). DDW issued two letters, dated January 7 and February 26, 2019, recommending approval of the Dual Plumbed Report and the plumbing design plans. DDW also provided a letter on December 22, 2021, that included recommended provisions for inclusion in this Notice for the Facility operation and recycled water uses.

Based on the information presented in the Title 22 Engineering and Dual Plumbed Reports, the wastewater and stormwater treatment system and proposed discharges are consistent with the General Order's requirements.

The General Order contains requirements that apply to all wastewater treatment systems and requirements that apply only to specific types of treatment and/or disposal systems. Applicable requirements and effluent limits are discussed in section 3 and summarized in Table 3 of this Notice. The General Order contains additional requirements that are not summarized in Table 3. The Discharger is responsible for all applicable requirements in the General Order, this Notice, and the Title 22 Engineering Report, as described in further detail below.

A copy of the General Order is enclosed with this Notice (Attachment C) and is accessible online.[2]

1. Facility and Discharge Description

NASA is the landowner and issues construction permits. Google leases from NASA and is responsible for the construction, operation, and maintenance of all infrastructure within the project boundaries on behalf of NASA. The Google Bay View Campus consists of three, three-level office buildings (G1, G2 and G3) and includes offices and general workspace, main kitchens, dining areas, various micro kitchens, courtyards, common spaces, conference rooms and/or events center. Future campus developments may include short-term employee housing and a parking structure. The site location, campus layout, and project boundaries are shown in Notice Attachment A, which is made part of this Notice by reference.

Stormwater is collected for reuse from the site surface runoff and collected in the stormwater pond system consisting of multiple ponds for raw stormwater detention, biofiltration, and retention. Rainwater collected from the G1, G2, and G3 building canopies will be combined with stormwater from the site and conveyed via overland surface flow as well as a piped conveyance system. Onsite overland stormwater flow is directed to the 704,605-gallon open water stormwater management pond, whereas stormwater runoff collected and pumped from storm drains will discharge to the onsite 1,324,127-gallon retention pond (also referred to as retention basin). The open water pond overflows to the retention pond. The combined stormwater is then pumped from the retention pond to an onsite treatment system. A pond water quality management system includes recirculation pumping through a biofiltration cell and aeration via diffusers.[3] If a rainfall event occurs and the ponds are at capacity, the retention pond will overflow to the detention pond and then to the existing Victory Road

[2] General Order: https://www.waterboards.ca.gov/board_decisions/adopted_orders/water_quality/2014/wqo2014_0153_dwq.pdf

[3] The open water pond area is 49,886 square feet (ft^2) with a maximum depth of six feet. The retention pond area is 39,417 ft^2 with a maximum depth of 5.5 feet. The areas of biofiltration cells 1 and 2 are 13,798 and 2,139 ft^2, respectively.

culvert to the western diked marsh located north of the site. Attachment A includes a schematic and figure of the stormwater pond system layout.

Only stormwater from the retention pond can be pumped to the central utility plant for treatment via an ultrafiltration membrane system with a membrane pore size of 0.03 micrometer (µm) designed for a surficial loading rate of 25 gallons per square foot per day. The stormwater treatment system is designed to treat up to 50,000 gallons per day (gpd) and consists of ozonation, ultraviolet (UV) disinfection, and activated carbon. The stormwater treatment system includes a single UV unit (with two lamps) to deliver an estimated dose greater than 100 millijoules / square centimeter (mJ/cm^2), which is the same UV system used for wastewater disinfection. Backwash from the system is directed to the onsite sewer collection system. If the system is not achieving turbidity targets, a coagulant can be injected directly into the influent to bind and settle out organic material. If it is determined that a coagulant is needed, a serpentine pipe contactor and static mixer will be used to ensure that the coagulant thoroughly mixes in the water before it is filtered out.

An onsite domestic wastewater collection system and lift station will transfer wastewater to an onsite wastewater treatment plant with a maximum design capacity of 85,000 gpd. When fully occupied (5,300 people), the maximum wastewater flow is estimated at 65,800 gpd. The wastewater treatment system has been designed to accommodate sanitary wastewater from NASA's collection system, which may be added to the wastewater flows being recycled in the future following consultation with the Water Board.

Domestic wastewater will be screened via a rotary drum 3-millimeter opening screen, pumped to two settling tanks totaling 26,000 gallons, equalized in six 10,000-gallon tanks with a combined capacity of 60,000 gallons, and treated through a moving bed bioreactor process followed by polishing through a horizontal flow subsurface treatment wetland. Screenings will be washed, compacted, bagged, and collected in a roll-off bin and hauled offsite. The moving bed bioreactor aerobic biological treatment system consists of a pre-anoxic zone, blower-aerated biochemical oxygen demand and nitrification zones, and clarification. A recycle pump will return a portion of the nitrified effluent to the beginning of the treatment process or pre-anoxic zone for denitrification. Waste activated sludge solids from the clarified zone are pumped to sludge holding tanks and hauled off-site. The five horizontal flow subsurface wetland treatment cells are a self-contained, lined, engineered system with no proposed discharges to the environment. The wetland cells have a synthetic liner to prevent infiltration or exfiltration to protect groundwater and surface water. The cells will be planted with vegetation species typically found in local wetlands. The wetland treatment cells effluent pump station and downstream processes at the central utility plant have been designed to manage the modeled peak rain event.

Following the treatment wetland, the effluent is pumped back to the central utility plant for final treatment via a Schreiber's Fuzzy Filter compressible media filter, Aquionics In-line 400+ UV disinfection system UV disinfection unit with two reactors, granular activated carbon adsorption filters for the removal of trace organics, and ozonation primarily for color removal and incidental disinfection. The media filter is sized to process the peak flowrate with a surface loading rate of 40 gallons per minute per square foot (gpm/ft^2). If needed, the treatment process could use a coagulant to achieve turbidity targets. The flow path will be in series through two UV units, each rated for supplying a dose of 100 mJ/cm^2. An onsite UV disinfection system spot-check bioassay test is not required at this time since the UV disinfection system will be operated to provide a double UV dose (200 mJ/cm^2). The ozone system includes a dissolved ozone sensor, an oxidation reduction potential (ORP) monitor,

and an ozone destruct system with carulite-200 media to catalyze any excess ozone back to oxygen.

The treated domestic effluent (recycled water) will be combined with the treated stormwater and rainwater in a 60,000-gallon nominal capacity blend tank equipped with a mixer prior to reuse in the central utility plant. The recycled water will be dosed with 12.5% sodium hypochlorite prior to storage to maintain an adequate residual chlorine concentration throughout the recycled water distribution system. Attachment A contains stormwater and wastewater treatment system process flow schematics and design drawings.

Stormwater, treated domestic wastewater, and potable water supplied by NASA will supply the Facility's non-potable water blending tank. The backup supply of NASA potable water will be provided to the blending tank via an air gap. The building is dual plumbed to allow for the use of potable and non-potable water sources. Domestic wastewater from toilet flushing, sinks, showers from the office, fitness, laundry, event, housing sources, and the food preparation areas will be collected and treated onsite to disinfected tertiary recycled water standards in accordance with 22 CCR sections 60301.230 and 60301.320. Disinfected tertiary treatment requirements are also referenced in section 3 of this Notice. Oil/water separators are installed on the kitchen wastewater drains to control contribution of fats, oils, and greases to the collection system. The tertiary treated recycled water will be blended with collected and treated stormwater and used to supply flushing fixtures (toilets and urinals), landscape irrigation, building and hardscape wash down, and cooling tower makeup within the central utility plant. Table 1 indicates estimated flow rates of recycled water to each use at design occupancy. The irrigation control system responds to weather tracking software and provides automatic irrigation scheduling based on collected weather data. The cooling towers will be used during peak cooling months to supplement the geothermal heat exchange piles or geopile system. The cooling towers are equipped with drift eliminators, and an oxidizing (bromine) and nonoxidizing biocide will be applied to the recirculating water to minimize the growth of Legionella and other microorganisms.

Table 1: Estimated Recycled Water Flow Rates per Use

Recycled Water Use	**Annual Average** (gallons/day)	**Maximum Day** (gallons/day)
Buildings (toilets + interior irrigation)	9,400	16,700
Outdoor irrigation	23,000	52,500
Cooling towers	1,100	5,800
Export off-site, building & hardscape wash down	44,700	63,200

2. California Environmental Quality Act

Enrollment of the Google Bay View Campus wastewater and stormwater recycled water treatment system under the General Order is categorically exempt from the California Environmental Quality Act (CEQA) CCR Title 14, section 15303. The Class 3 categorical exemption applies to projects consisting of the construction and location of limited numbers of new, small facilities or structures, the installation of small new equipment and facilities in small structures, and the related conversions of existing small structures, and includes utility improvements and extensions. The onsite recycled water treatment system, as described in the Facility and Discharge Description of this Notice, meets these criteria. As a result, the Water Board filed a Notice of Exemption to document the proposed action. The Water Board action also consists of enrolling the Google Bay View Campus wastewater and stormwater

recycled water treatment system and associated discharges under the General Order, which applies to small wastewater treatment systems with an average monthly flow of less than 100,000 gpd.

3. Site-Specific Requirements and Effluent Limitations

The General Order contains prohibitions and specifications that apply to all wastewater systems, and specifications that apply only to specific types of treatment and/or disposal systems. **Table 3** of this Notice provides a summary of applicable discharge requirements for the subject system, both general and specific, and associated effluent limits. In order to maintain coverage under the General Order, the Discharger is required to operate consistently with its approved Title 22 Engineering Report; O&M and Contingency Plan; conditions of DDW's approval letter (Notice section G & Attachment E); 22 CCR Uniform Statewide Recycling Criteria; the requirements set forth below; and the Monitoring and Reporting Program in Notice Attachment B.

Treatment system managers and operators must be familiar with the applicable requirements contained in the General Order, this Notice, and the Facility's Title 22 Engineering Report. 23 CCR sections 3680.1, *Grade Levels of Operators at Privately Owned Wastewater Treatment Plants*, and 3680, *Grade Levels of Operator Certification*, state the operator grade level requirements according to the wastewater treatment plant classification. Wastewater treatment plant classification criteria are defined in 23 CCR section 3675. The Facility aligns with a Class III wastewater treatment plant classification since it will provide tertiary treatment for a design flow of less than 1.0 million gallons day. For a Class III wastewater treatment plant, the Chief Plant Operator's minimum certification level is III and the Designated Operator-in-Charge must meet a minimum grade level of II. The State Water Board reviews the wastewater plant classification form, makes the final determination, and sends a letter indicating the classification and appropriate operator grades. Contact the State Water Board Plant Classifications Analyst via email to wwopcertprogram@waterboards.ca.gov to confirm the classification. Additional information on the wastewater treatment plant classification process is accessible online.[4]

A. Effluent Limitations

The wastewater treatment system is required to be operated and maintained to achieve the design objectives included in the Title 22 Engineering Report and to demonstrate compliance with technology-based performance effluent limits for biochemical oxygen demand and total suspended solids. The Discharger shall ensure that the recycled water meets the following disinfected tertiary recycled water quality standards for unrestricted use established in 22 CCR sections 60301.230 and 60301.320.

1. The turbidity of the filtered wastewater shall not exceed any of the following:
   a. An average of 2 NTU within a 24-hour period;
   b. 5 NTU more than 5 percent of the time within a 24-hour period; and
   c. 10 NTU at any time.

2. Total coliform measured in the disinfected effluent does not exceed the following:

[4] Wastewater treatment plant classification: https://www.waterboards.ca.gov/water_issues/programs/operator_certification/wwtp.html

a. 2.2 MPN / 100 mL (7-day median); and

b. 23 MPN / 100 mL (not more than one sample in 30 days); and

c. 240 MPN / 100 mL (maximum any sample).

The effluent limits in Table 2 apply to the treated domestic wastewater distributed for recycled water uses. The effluent limits are also referenced in Table 3 of this Notice.

Table 2: Treated Domestic Wastewater (Effluent) Limitations

Constituent	Units	Limit
Biochemical Oxygen Demand (BOD)	mg/L	30 (monthly average)
		45 (7-day average)
Total Suspended Solids	mg/L	30 (monthly average)
		45 (7-day average)
Turbidity	NTU	Average of 2 NTU within a 24-hour period
		5 NTU more than 5 percent of the time within a 24-hour period
		10 NTU maximum, any time
Total Coliform Bacteria	MPN / 100 mL	2.2 MPN / 100 mL rolling 7-day median
		23 MPN / 100 mL 30-day maximum, if more than 1 exceedance occurs during 30-day period. Can exceed maximum limit once.
		240 MPN / 100 mL maximum, any time

Staff evaluated the need for a nitrogen effluent limit using the method contained in Attachment 1 of the General Order and determined a nitrogen effluent limit is not applicable at this time considering that the recycled water will be applied to land via irrigation based on planting hydrozone requirements, soil percolation capacity, and soil moisture sensors to prevent overwatering. Due to the Facility's proximity to San Francisco Bay and a regional effort to monitor and assess nutrient loadings to the Bay, influent and effluent monitoring for total nitrogen is included in the attached Monitoring Program. The Monitoring Program also requires the calculation of the wastewater treatment system's total nitrogen percent reduction. The nitrogen effluent limit applies to discharges to land greater than 20,000 gpd. At design capacity, the Discharger estimated an annual average flow to outdoor irrigation of 23,000 gpd. The Water Board may reevaluate the nitrogen effluent limit if the average discharge to land flow rate exceeds 20,000 gpd and the recycled water nitrogen concentrations and treatment system nitrogen reduction effectiveness warrants a reassessment or if climate change impacts result in groundwater rise.

B. Sampling and Analysis

1. Disinfected tertiary recycled water shall be sampled at least once daily during recycled water production for total coliform bacteria. The samples shall be taken from the disinfected effluent and shall be analyzed by an approved laboratory as defined by 22 CCR section 60301.100.

2. Per 22 CCR section 60321, disinfected tertiary recycled water shall be continuously sampled for turbidity using a continuous turbidity meter and recorded following filtration. Compliance with the daily average operating filter effluent turbidity (average of 2 NTU within a 24-hour period) shall be determined by averaging the levels of recorded turbidity taken at four-hour intervals over a 24-hour period. Compliance with the filtered effluent turbidity limit pursuant to 22 CCR section 60301.320(a)(2)(B) (5 NTU more than 5% of the time within a 24-hour period) shall be determined using the levels of recorded turbidity taken at intervals of no more than 1.2-hours over a 24-hour period. Should the continuous turbidity meter and recorder fail, grab sampling at a minimum frequency of 1.2-hours may be substituted for a period of up to 24 hours.

C. Bypass

1. There shall be no bypassing of untreated or partially treated wastewater from the recycled water treatment system or any intermediate unit process to the point of use.

2. All unit processes must be well maintained and in operation at all times consistent with the original design and operational intent.

D. Operating Records and Reports

1. Operating records shall be maintained at the Facility or a central depository within the Discharger's operating agency. These shall include: all analyses specified in the reclamation criteria; records of operational problems, plant and equipment breakdowns, and diversions to emergency storage or disposal; all corrective or preventive action taken.

2. Process or equipment failures triggering an alarm shall be recorded and maintained as a separate record file. The recorded information shall include the time and cause of failure and corrective action taken.

3. The Discharger is responsible for ensuring the Title 22 Engineering Report on file with the Water Board and DDW remains current and accurate.

E. Notification

1. The Discharger shall notify the Water Board of recycled water spills or unauthorized releases in accordance with the Water Board's recycled water spill notification protocol (Notice Attachment D).

2. The Discharger shall notify the Water Board, DDW, and the Santa Clara County Environmental Health Department of the following:

   a. Any incidence of backflow from the dual-plumbed recycled water system into the potable water system within 24 hours of the discovery of the incident.

   b. Any discharge of untreated or partially treated wastewater to the use area, and the cessation of same (22 CCR section 60329(d)).

The current Water Board and DDW notification contacts are:

Water Board contacts:	Melissa.Gunter@waterboards.ca.gov Margaret.Monahan@waterboards.ca.gov WDR.Monitoring@waterboards.ca.gov
DDW contacts:	Van.Tsang@waterboards.ca.gov Eric.Lacy@waterboards.ca.gov
Santa Clara County:	(408) 918-3400

F. Cross-Connection Control

Since the Facility is located within the NASA building jurisdiction, NASA will inspect the installation of the recycled water piping system for proper labeling and separation and oversee a cross-connection shutdown test at start-up of the recycled water system. Conditions per DDW's letter dated February 26, 2019, recommending approval of the Dual Plumbed Report and plumbing design plans, include:

1. Prior to the initial operation of the recycled water system and annually thereafter, the Discharger shall ensure that the dual plumbed system within buildings G1, G2, and G3, central utility plant, and use area are inspected for possible cross-connections with the potable water system.

2. The recycled water system shall also be tested for possible cross-connections at least once every four years. The testing shall be conducted in accordance with the method described in the report submitted pursuant to 22 CCR section 60314. The inspections and the testing shall be performed by a cross-connection control specialist certified by the California-Nevada section of the American Water Works Association or an organization with equivalent certification requirements.

3. A cross-connection shutdown test procedure shall be submitted to DDW and NASA for review and approval prior to the test.

4. A written report documenting the result of the cross-connection inspection or testing for the prior year shall be submitted to the Water Board and DDW within **30 days** following completion of the inspection or testing.

5. Any backflow prevention device installed to protect the public water system serving the dual plumbed recycled water system must be inspected and maintained in accordance with 17 CCR section 7605.

G. Division of Drinking Water Conditions

DDW provided a letter on December 22, 2021, with recommended provisions on the Discharger's production, operation, and use of recycled water, from which are the following conditions:

1. The Discharger must comply with all applicable requirements set forth in Titles 17 and 22 of the CCR for the production, distribution, and use of recycled water.

2. All recycled water produced at the Facility must be filtered and disinfected using the Schreiber Fuzzy Filter compressible media filter and Aquionics In-line 400+ UV disinfection system, respectively. Changes, additions, or modifications to the filter or UV disinfection system must be approved by DDW and the Water Board prior to implementation.

3. The Discharger shall provide a sufficient number of qualified personnel to operate the Facility effectively as to achieve the required level of treatment at all times. Qualified personnel must be those meeting requirements of the California Water Code (Water Code) chapter 9 (commencing with section 13625).

4. The surface loading rate onto the Schreiber Fuzzy Filter compressible media filter shall not exceed 40 gallons per minute per square foot.

5. The maximum flow rate through the Schreiber Fuzzy Filter compressible media filter shall not exceed 90 gallons per minute.

6. The filter bed shall be comprised of at least 30-inches of uncompressed filter media.

7. The Discharger shall implement a response action per the O&M and Contingency Plan when filter influent turbidity concentrations exceed the threshold levels of 10 NTU more than five percent of the time within a 24-hour period or 15 NTU any time.

8. In lieu of an UV disinfection system spot check bioassay, the UV system shall be operated with two Aquionics In-line 400+ UV reactors in series, each to deliver a minimum UV dose of 100 mJ/cm^2, such that the water receives a minimum UV dose of 200 mJ/cm^2 at all times.

9. On-line monitoring of flow, UV intensity, UV transmittance, and power shall be conducted at all times.

10. Flow meters, UV intensity sensors, and UV transmittance monitors shall be calibrated to ensure proper disinfection.

11. At least monthly, all duty UV intensity sensors shall be checked for calibration against a reference UV intensity sensor or laboratory measurement.

12. For all UV intensity sensors in use, the ratio of the duty UV sensor intensity to the reference UV sensor intensity shall be less than or equal to 1.2. If the calibration ratio is greater than 1.2, the failed duty UV sensor must be replaced by a properly calibrated sensor and recalibrated by a qualified facility. The reference UV intensity sensors shall be recalibrated at least annually by a qualified facility using a National Institute of Standards and Technology traceable standard.

13. The UV transmittance meter shall be inspected and checked against a reference bench-top unit or portable meter weekly to document accuracy. The weekly sampling frequency may be reduced based upon data demonstrating the instrument's drift frequency.

14. If the on-line analyzer UV transmittance reading varies from the bench-top spectrophotometer or field meter UV transmittance reading by 2% or more, the on-line UV transmittance analyzer shall be recalibrated by a procedure and frequency recommended by the manufacturer.

15. Flow meters measuring the flow through a UV reactor must be verified to determine accuracy at least monthly via checking the flow reading against other flow determination methods.

16. The Facility shall be designed with built-in automatic reliability features that must be triggered by critical alarm setpoints.

17. Conditions that should divert effluent to waste include the following:
    a. UV dose below 100 mJ/cm^2 for either inline 400+ UV reactor
    b. UV transmittance is below 55%

18. The Facility shall be operated in accordance with the approved O&M and Contingency Plan, which specifies the operational limits and responses required for critical alarms. A copy of the O&M and Contingency Plan shall be maintained at the Facility and be readily available to operations personnel and regulatory agencies. A quick reference plant operations data sheet must be posted at the Facility and include the following information:
    a. Alarm set points for UV dose and UV transmittance.
    b. Values of UV dose and UV transmittance when effluent must be diverted to waste.
    c. Required frequency of verification and calibration for all meters/analyzers measuring flow, UV intensity, and UV transmittance.
    d. Required frequency of mechanical cleaning and equipment inspection.
    e. UV lamp hour tracking procedures and replacement intervals.

19. Equivalent or substitutions of equipment shall be approved by DDW and the Water Board prior to implementation with an adequate demonstration of equivalent disinfection performance.

20. The Discharger shall sample the disinfected tertiary recycled water at least once daily during recycled water production for total coliform bacteria in accordance with the Monitoring and Reporting Program (Monitoring Program) in Notice Attachment B. The samples shall be taken after the UV disinfection system but prior to the granular activated carbon (GAC) bed and chlorine injection point and analyzed by an Environmental Laboratory Accreditation Program (ELAP)-accredited laboratory.

21. The Discharger shall sample the disinfected tertiary recycled water for turbidity using a continuous turbidity meter and recorder after the Schreiber Fuzzy Filter compressible media filter on the filtrate effluent pipe prior to the 550-gallon polyethylene lift station. The results of the daily average turbidity determinations shall be reported to the Water Board in accordance with the Monitoring Program (Attachment B).

22. A preventive maintenance program shall be provided at the Facility to ensure that all equipment is kept in a reliable operating condition.

23. Although no domestic water supply wells are currently proposed within the use area, no irrigation with disinfected tertiary recycled water shall occur within 50 feet of any

domestic water supply well unless all of the conditions listed in 22 CCR section 60310(a) have been met.

24. No impoundment of disinfected tertiary recycled water may occur within 100 feet of any domestic water supply well.

25. Any use of recycled water must comply with the following:
    a. Any irrigation runoff must be confined to the recycled water use area, unless the runoff does not pose a public health threat and is authorized by the regulatory agency.
    b. Spray, mist, or runoff must not enter dwellings, designated outdoor eating areas, or food handling facilities.
    c. Drinking water fountains must be protected against contact with recycled water spray, mist, or runoff.

26. All use areas where recycled water is used that are accessible to the public shall be posted with signs that are visible to the public, in a size no less than 4 inches high by 8 inches wide, that include the following wording: "RECYCLED WATER - DO NOT DRINK". Each sign must display an international symbol similar to that shown in 22 CCR section 60310 (figure 60310-A). DDW may accept alternative signage and wording, or an educational program, provided the applicant demonstrates to DDW that the alternative approach will assure an equivalent degree of public notification.

27. No physical connection can be made or allowed to exist between any recycled water system and any separate system conveying potable water.

28. The recycled water piping system that are in areas subject to access by the general public shall not include any hose bibs. Only quick couplers that differ from those used on the potable water system can be used on the portions of the recycled water piping system in areas subject to public access.

H. Operating Conditions

1. The 5-log virus inactivation (99.999% removal) required to meet the 22 CCR section 60301.230 disinfected tertiary recycled water requirement shall be achieved.
2. If the Discharger conducts a spot-check bioassay test to establish site-specific UV operating conditions in support of a reduction in the double UV dose requirement, then the following conditions apply:
    a. The spot-check bioassay test procedures or protocols shall be submitted to DDW for review and approval prior to initiating the test. A copy of the procedures shall be submitted to the Water Board.
    b. The spot-check bioassay test shall be conducted in accordance with the National Water Research Institute *UV Disinfection Guidelines for Drinking Water and Water Reuse*.
    c. Operating conditions for the UV disinfection system based on the spot-check bioassay test results will be provided by DDW after the test has been completed and the test report submitted for review.

3. The wetland and the stormwater pond systems shall be managed to control the breeding of flies, mosquitoes, and other vectors of public health significance.

4. The Discharger shall discontinue the distribution of recycled water to the indoor uses during any period in which it has a reason to believe that the quality of the recycled water is not meeting the Title 22 Uniform Statewide Recycling Criteria specifications.

I. Engineering Report Updates

1. Should the Discharger consider other uses of recycled water or expansion of recycled water to additional facilities, an updated Title 22 Engineering Report shall be submitted to the Water Board and DDW for review and approval.
2. The Discharger is responsible for assuring the Title 22 Engineering Report on file with the Water Board and DDW remains current and accurate and addresses comments provided by both agencies during the permitting process.

J. Blended Water Storage Tank Conditions

The following conditions are applicable to the final blend tank to ensure safe storage of the blended recycled water for later use:

1. The storage tank must be located at least 25 feet from a flowing surface water body.
2. Ensure there is proper screening and sealing of the tank to prevent the entry or egress of insects and mosquitos. The Environmental Protection Agency recommends a minimum of #24 mesh for vent screens.
3. Reduce the potential for microbial growth in the recycled water distribution system and storage tanks by avoiding stagnation resulting from dead zones or prolonged periods of zero-flow or low flow that create long residence times.
4. Any storage tank containing recycled water for reuse applications shall be managed in a manner to control odor or nuisance conditions.

K. Reports to be Submitted to the Water Board

1. The Discharger shall notify the Water Board following commencement of the operation of the Facility resulting in recycled water uses. This notification shall be submitted no later than **5 calendar days** after the Facility is operational.

2. The Discharger shall submit complete plan-view design drawings of the completed wastewater system and all its components to the Water Board if changes made during construction are not included in the plans previously submitted to the Water Board. The plans shall be submitted no later than **30 calendar days** after finalization of the as-built plans.

In accordance with General Order Provisions E.1.a, b, and c, the following technical reports, shall be submitted as described below and in the referenced Order Provisions and shall be acceptable to the Water Board's Executive Officer.

The objective of these reports is to provide appropriate and current information about how to operate and maintain the treatment system in order to assure adequate and reliable wastewater management for the life of the Facility, and ongoing compliance with waste discharge requirements.

The requirements herein address initial report submittals. As time proceeds, and the wastewater treatment system matures, it is anticipated that repairs, replacements, improvements, or operational changes will occur, and these will need to be incorporated into the operating reports accordingly. Whenever any of these reports are updated, the Discharger shall submit the updated sections or revised report to the Water Board.

3. **Within 90 days** of the issuance of this Notice, **by not later than October 25, 2022**, the Discharger shall submit a **Spill Prevention and Emergency Response Plan** (Response Plan) consistent with the requirements of General Order Provision E.1.a. Some of the Response Plan content requirements (e.g., operation and control of wastewater treatment and collection system maintenance) have been addressed in the O&M and Contingency Plan. The O&M and Contingency Plan section can be referenced for those topics. The Response Plan shall, at a minimum, address the following:
   a. Solids handling – The Sludge Management Plan per Notice K.5. can be submitted or referenced to satisfy this requirement
   b. Emergency response
   c. Recycled water spill notification procedures per the Water Board spill reporting protocol included in Notice Attachment D
   d. In the event of a wastewater collection system overflow
   e. Operation and control of wastewater treatment
   f. Collection system maintenance
   g. Notification procedures

4. **Within 90 days** of the issuance of this Notice, **by not later than October 25, 2022**, the Discharger shall submit a **Sampling and Analysis Plan** consistent with the requirements of General Order Provision E.1.b.

5. **Within 90 days** of the issuance of this Notice, **by not later than DATE, 2022**, the Discharger shall submit a **Sludge Management Plan** consistent with the requirements of General Order Provision E.1.c.

4. Monitoring and Reporting Program

The Discharger shall comply with the Monitoring and Reporting Program in Attachment B, which is attached hereto and made part of this Notice by reference.

5. Enforcement

Please review this Notice carefully to ensure that it completely and accurately reflects the wastewater and stormwater treatment systems and discharge. If the Discharger violates the terms or conditions listed, the Water Board may take enforcement action, including assessment of administrative civil liability, as authorized by provisions of the Water Code. Prior to allowing changes to the wastewater strength or generation rate, or if the method of waste discharge changes from that described in the Title 22 Engineering Report, the Discharger shall contact the Water Board to determine if submittal of an updated Report of Waste Discharge or Title 22 Engineering Report is required.

The Discharger shall manage and discharge the recycled water subject to the terms and conditions of the General Order, this Notice, and the Title 22 Engineering Report and will

maintain exclusive control over the discharges. As such, the Discharger is responsible for compliance with the General Order and this Notice.

6. Documents

For reference purposes, the documents discussed above include the following:

1. California Department of Health Services Fuzzy Filter compressible media acceptance letter dated February 24, 2003. Filed in the Water Board's electronic database with the document handle number (ECM No.) 9405190
2. Division of Drinking Water's Dual Plumbed Report acceptance letter dated January 7, 2019. ECM No. 9405191
3. Division of Drinking Water's Dual Plumbed Report revised acceptance letter dated February 26, 2019. ECM No. 9405192
4. Google Title 22 Engineering Report for the Treatment and Reuse of Wastewater and Stormwater dated December 15, 2021. ECM No. 9349724
5. Google Title 22 Technical Report: Preliminary Operations and Maintenance Plan and Contingency Plan dated December 15, 2021. ECM No. 9349721
   O&M Enclosures B and C. ECM No. 9349717
6. Division of Drinking Water's Title 22 Engineering Report conditional approval letter dated December 22, 2021. ECM No. 9405193
7. Google Report of Waste Discharge submitted January 21, 2022. ECM No. 9405194

7. Annual Fees

The Discharger is responsible for payment of an annual fee, pursuant to Water Code section 13260 and the Annual Fee Schedule as stated in CCR Title 23, division 3, chapter 9, section 2200. The fee amount is determined by the type of order and the discharge's threat to water quality and complexity rating. An application fee of $7,486, which will also be the annual fee, is due to the Water Board. This represents the fee for a discharge with a threat and complexity rating of 3-B. The Water Quality Fee Schedule is subject to change annually and is accessible online under Waste Discharge Requirements Fee Schedule. The State Water Board's Fees webpage contains additional information on paying fees.[5] The Waste Discharger ID (WDID) is associated with billing **2 43GWDR440668**.

The fee is due and payable on an annual basis until coverage under the General Order is formally rescinded. The Discharger must provide written notice if the wastewater discharge ceases so that we may terminate coverage. To terminate coverage voluntarily, the Discharger must submit in writing a Notice of Termination that includes, at a minimum, the Discharger's name and contact information, Facility name and address, and reason for requesting termination of coverage. However, termination shall not be considered effective until the Water Board provides written approval of the Notice of Termination.

Authorization for Coverage under Order WQ 2014-0153-DWQ

I hereby approve the Google Bay View Campus Wastewater and Stormwater Treatment System as described in the Report of Waste Discharge and Title 22 Engineering Report, and as approved by the Division of Drinking Water's letters, and the Water Board herein, to

[5] CCR Fee Schedule: http://www.waterboards.ca.gov/resources/fees/water_quality/#wdr
State Water Board Fees webpage: https://www.waterboards.ca.gov/resources/fees/

henceforth be regulated under the General Order. Failure to abide by the conditions of the General Order, this Notice, the Monitoring and Reporting Program, and the Title 22 Engineering Report, could result in enforcement actions, as authorized by provisions of the Water Code.

If you have any questions or comments, please contact Melissa Gunter at (510) 622-2390 or Melissa.Gunter@waterboards.ca.gov.

Sincerely,

for

Eileen White
Executive Officer

Table: Table 3: Summary of Applicable General Order Requirements

Attachments:
A. Figures: Site Location & Site Layout
Water Management Schematics
Treatment Process Flow Diagrams
B. Monitoring and Reporting Program
C. Water Quality Order WQ 2014-0153-DWQ
D. Recycled Water Spill Notification Protocol
E. Division of Drinking Water's dual plumbing and engineering report letters

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TABLE 3: SUMMARY OF APPLICABLE GENERAL ORDER REQUIREMENTS

General Order Section	Page	Numeric Limit [6]	**Requirement** (refer to General Order and Notice for details)
A. PROHIBITIONS			
A.1 - A.11	15	--	Prohibitions
B. REQUIREMENTS BY WASTEWATER SYSTEM TYPE			
B.1.a - B.1.l	16	• 25-foot setback from land application areas and flowing stream • 50-foot setback from treatment system and flowing stream	All Wastewater Systems, which includes the setback distances for Land Application Area (**disinfected tertiary recycled water**) in General Order Table 3.
B.1.a	16	Discharger shall not accept wastewater in excess of the wastewater treatment system's design capacity	85,000 gallons per day domestic wastewater treatment system design flow limit
B.3.a - B.3.g	21	--	Aerobic Treatment Units - Refer to General Order sections
B.4.a - B.4.b	23	--	Activated Sludge Systems - Refer to General Order sections
B.6.a - B.6.h	24	--	Subsurface Disposal Systems - Refer to General Order sections
B.7.a - B.7.h	26	--	Land Application and/or Recycled Water Systems – Refer to General Order sections
B.8.a - B.8.f	26	--	Sludge/Solids/Biosolids Disposal - Refer to General Order sections
C. GROUNDWATER AND SURFACE WATER LIMITATIONS			
C.1	27	--	Refer to General Order section
D. EFFLUENT LIMITATIONS			
Activated Sludge, Membrane Bioreactor (MBR), or similar			
D.1.a	27	30 mg/L (monthly avg) 45 mg/L (7-day average)	Five-day Biochemical Oxygen Demand (BOD_5) (Refer to General Order Table 4 for averaging allowances)
D.1.a	27	30 mg/L (monthly avg) 45 mg/L (7-day average)	Total Suspended Solids (TSS) (Refer to General Order Table 4 for averaging allowances)

[6] MPN / 100 mL denotes most probable number per 100 milliliters; NTU denotes nephelometric turbidity units

TABLE 3: SUMMARY OF APPLICABLE GENERAL ORDER REQUIREMENTS

General Order Section	Page	Numeric Limit [6]	**Requirement** (refer to General Order and Notice for details)
D.1.a	27	Nitrogen effluent limit is not applicable at this time	Influent and effluent nitrogen monitoring included to determine domestic wastewater treatment system's percent nitrogen reduction.
California Code of Regulations Title 22 Water Recycling Criteria Requirements			
B.7.f	26	a. 2.2 MPN / 100 mL (7-day median); & b. 23 MPN / 100 mL (not more than one sample in 30 days); & c. 240 MPN / 100 mL (maximum any sample)	Total coliform bacteria: Disinfected tertiary recycled water total coliform requirements per California Code of Regulations Title 22 (22 CCR) section 60301.230. If recycled water is applied to land surface or flushing fixtures, it shall comply with the Title 22 water recycling criteria.
B.7.f	26	a. 2 NTU average within a 24-hour period b. 5 NTU more than 5% of the time within a 24-hour period; & c. 10 NTU maximum any time	Turbidity: Disinfected tertiary recycled water turbidity requirements following media filtration applicable to the domestic wastewater treatment system per 22 CCR section 60301.320.
E. PROVISIONS			
Technical Report Preparation Requirements			
E.1.a	28	Submit by **October 25, 2022**, to Water Board	Spill Prevention and Emergency Response Plan
E.1.b	29	Submit by **October 25, 2022**, to Water Board	Sampling and Analysis Plan
E.1.c	30	Submit by **October 25, 2022**, to Water Board	Sludge Management Plan
E.2. all sections	30	--	Entire section provisions applicable to all wastewater treatment systems
E.2.h	32	Have onsite	Copy of the General Order, Notice, and Title 22 Engineering Report
E.2.k	32	--	Certified wastewater treatment operator. Refer to **Notice section 3.** Contact the State Water Board Plant Classifications Analyst at wwopcertprogram@waterboards.ca.gov
E.3.	34	--	General Reporting Requirements provisions applicable to all wastewater treatment systems
E.4.	35	--	Monitoring Requirements provisions applicable to all wastewater treatment systems